Multimedia Tools and Applications (2023) 82:1873–1904
https://doi.org/10.1007/s11042-022-13312-1

A comprehensive survey on image authentication for tamper detection with localization

H R Chennamma [1] · B Madhushree [1]

Received: 22 July 2021 / Revised: 17 January 2022 / Accepted: 30 May 2022 /
Published online: 14 June 2022
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract
The data that is stored or transmitted in digital form is susceptible to be attacked. Digital images are particularly vulnerable to unauthorized access and illicit alterations. As a result, image credibility verification approaches are constantly gaining popularity due to their importance in a variety of social concerns, including the government, military, forensics, and electronic commerce, among others. Even a minor attack could have disastrous consequences. Therefore, protecting images from tampering has become increasingly crucial. Many approaches have been developed in the literature to ensure the authenticity and integrity of digital images. These methodologies can be grouped into active and passive approaches for image authentication and tampering detection. Active image authentication has a general solution based on watermarking, signature and hybrid techniques. Passive authentication, on the other hand, relies on the image's inherent natural characteristics to provide implicit authentication. Digital image forensic tools have been utilized for identifying the signs of digital tampering. We have found from the literature of the past three decades that image credibility verification approaches have achieved promising results. However, accurate image tamper localization remains an unsolved problem and a fairly difficult task too, if it is performed carefully with expertise. This motivated us to present a comprehensive review of image authentication techniques, which enables not only image tamper detection but also localize the manipulated region. We also discussed the future work.

Keywords Digital signature · Digital watermarking · Explicit authentication · Image authentication · Implicit authentication · Tamper localization

1 Introduction

IMAGE authentication is a process of enabling an image to verify its authenticity and integrity. It has gained huge attention because of its importance in providing security and integrity. In this

✉ H R Chennamma
anuamruthesh@gmail.com

[1] Department of Computer Applications, JSS Science and Technology University, Mysuru, India

digital era, images are more prone to manipulations and it is very difficult to retain them intact. As a result, safeguarding confidential images such as medical images, military images, and so on is more critical. Integrity refers to a state or quality that is whole, intact, and has not been deliberately altered once it has been produced from its source. This is the concept of wholeness or the state of being genuine, or realness. Authentic [126] refers to the stated origin or authority that can be verified and is not copied. Because of its unique integrity criteria, image authentication differs from other types of data authentication. An image can often be seen in a variety of file formats, each of which can provide almost identical graphic details but completely different data representations. Images are compressed as well to reduce redundancy, but this has no effect on their semantics. In certain implementations, image authentication is needed to be robust and appropriate for image enhancement operations which do not change the image semantics (e.g., contrast correction, histogram equalization and lossy compression), whereas being sensitive to malicious attacks (inserting or modifying the details into/of the image).

Many survey papers have been published on forgery detection where each paper has different type of representation. Kaur Gill et al. gives the idea of different type of forgery techniques that include only passive techniques [55]. Categorization of forgery detection is important that was explained by Vinolin et al. It includes hierarchical structure of different forgery techniques of both active and passive but concentrated more on passive techniques [153]. Recently researchers uses deep learning algorithm for tamper detection and localization for better results so doing survey based on deep learning technique is more essential. Barad et al. gives review on image tampering detection using deep learning techniques where it tells about passive techniques, future directions and datasets [17].

It is quite straightforward to determine whether or not an image has been tampered with. However, locating the tampered region efficiently is a difficult task. We believe that efficient localization of the tampered region is nearly impossible when alterations are made by a skilled expert using sophisticated image editing tools. As a result, image authentication for tamper detection with localization may provide total protection for the image. Anyhow, image authentication can serve two purposes, such as verification of both authenticity and integrity of images. This article presents a review of the tamper localization problem based on active and passive image authentication approaches.

The rest of this paper is structured as follows: Section 2 describes state-of-the-art techniques for tamper detection with localization based on active image authentication, including watermark, signature and hybrid approaches. Section 3 presents state-of-the-art techniques for tamper detection with localization based on passive image authentication using pixel-based, format-based, camera-based, physical environment based and geometry based approaches. Section 4 describes publicly available datasets which were used in the literature. Section 5 contains the conclusion and future work.

2 Active approaches

Active image authentication is a type of authentication that uses a known authentication code that is either inserted in or transmitted along with the image to determine the reliability and credibility of the information and the sender. This type of approach, however, involves the unique generation of a signature or watermark at the time of recording or sending. Digital watermarking, digital signatures, and hybrid methods are the three forms of active approaches.

2.1 Digital watermarking

Digital watermarking is the mechanism by which digital information related to authentication is secretly embedded in the digital media/signal to be stored or transmitted. The embedded information is regarded as a digital watermark. The signal in which the watermark is embedded is called the host signal or carrier signal. The carrier signal may be audio, video, images or text. A watermark is embedded in the carrier signal such that only an intended receiver can extract it to verify the authenticity and integrity of the carrier signal. This usually takes place with pre-shared information, such as a secret key shared between the sender and the receiver. This mutual key is used to define, locate and extract the watermark as well as to authenticate the carrier signal [181]. Nowadays, the watermark embedded in the carrier signal can be recovered, even if it is altered by non-destructive attacks. The majority of watermarking research focuses on copyright protection. Many services, such as image authentication, content protection, as well as many fundamental concerns, are yet unresolved. Still, whether a fragile watermark, semi-fragile watermark and robust watermark should be employed for authentication is debatable [60, 122, 147, 169]. Anyhow, the main objective of watermarking is to conceal information in digital images in order to ensure security, as seen in Fig.1.

Integrity service is described by the security community as data that is delivered and received must be in the same format. However, it is not well adapted to digital images. Because image editing is unavoidable sometimes due to image enhancement, but semantics of the image cannot be changed. However, image content protection is the main concern of image authentication. First and foremost, while providing authentication services for still images, one should mainly focus on different non-malicious adjustments, such as format conversion, augmentation and compression. This, however, does not apply to all types of images. It mainly depends on the field of application and the type of images utilized. In terms of integrity, medical images and artistic images are not the same. The main objective of this paper is to demonstrate the different approaches to image authentication with tampering localization. Firstly digital watermarking technique is explained where it is divided into subsections based on the type of service it provides for image authentication and tampering localization.

2.1.1 Fragile watermarking

The majority of image authentication services rely primarily on fragile watermarks designed to protect copyright [173]. This technique mainly deals with inserting a particular watermark.

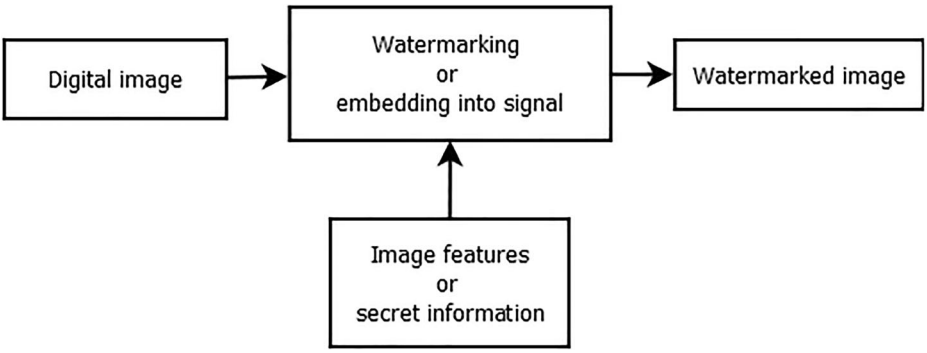

Fig. 1 Basic Watermarking Procedure

Therefore, if there is any alteration with the image content, it causes variations in the watermark. That is why the process of locating the watermark distortions is necessary to find the region of the image that is tampered with. The main disadvantage of this method is the inability to distinguish between harmful and non-harmful attacks. Wong et al. proposed a method which can authenticate the image and verify the ownership information using both visible and invisible watermarking techniques [165]. The result shows that if the digital key that is entered by the user is correct, then only the watermark can be extracted from the original image. To work with the frequency domain, Wu et al. present a novel method employed to authenticate the image using the look up table [LUP], which is prepared using the information gathered in the frequency domain [168]. Here, a fragile watermark is used for embedding. The watermark is embedded into the original image by considering the dominant features and this information is compressed. Features are extracted from the image and used to detect the tampered region. This method can detect the alternations and location of tampered regions in the image accurately. To enhance the process, Paquet et al. [124] proposed a unique method which is accurate at image tampering detection using fragile watermarking. Discrete Wavelet Transformations are used to embed the watermark into the image. It is good at spatial and frequency tampering localization without any prior information. This method helps to decrease the false negative rate. Lu et al. [100] proposed a novel method for creating a watermark that is both robust and fragile. This method helps with content protection as well as tamper detection. This novel method assures that any kind of attack that come across at least one technique will survive. This increases the security of the image. It is referred to as a multipurpose watermarking scheme that is resistant to all types of attacks. Authenticating and localizing Lena or bamboo images is common in most research papers, but this paper authenticates 3-D polygonal meshes [88], which are difficult to design. However, it formulizes the trade-off of the causality problem in Yeo and Yeung's method [172] for a new fragile watermarking scheme. This scheme can not only achieve localization of malicious modifications in visual inspection, but also is immune to certain incidental data processing.

An advanced fragile watermarking is proposed using LBP for tamper localization. Firstly, original image is divide into number of blocks the LBP is applied to generate binary image then Arnold transform is applied to obtain binary watermark. Then binary watermark is embedded into LSB. To detect the forged area exclusive-OR operation is applied to extract the watermark from watermarked image. If the watermark is extracted properly then the image is not tampered if it not then the image is tampered [179]. For real time application tamper localization is needed, Azeroual et al. proposed a technique where it detects the tampered area in real time using Faber-Schauder- DWT. Experimental result shows that FSDWT achieves good result with simple operations and has low complexity [15]. For effective Self-recovery of tampered area author gracia et al. proposed fragile watermarking technique to detect and recover the tampered area [111]. To improve the recovery of tampered area Abdelhakim et al. proposed k-means clustering. This fragile watermarking employed under spatial domain to embed the watermark into the image. K-means clustering is used to recover the tampered area [3].

a) Check sum

As the first and most widely used method for detecting image tampering, the check-sum was added to the Least Significant Bit [LSB] of image pixels. The process of verification is almost the same as embedding. This involves comparing the check-sum of the original image with the

check-sum of the tested image obtained from the LSB. The main advantage of this approach is that there will be no variation in image visibility, and it is effective at detecting tampering. For instance, if two pixels of an image block are interchanged, then the checksum also gets changed. That is due to every pixel in the block being multiplied by a distinct co-efficient. The function of the co-efficient and the image pixel are dependent on blocks, because there is a possibility to replace or interchange the whole block without any unrecognized alterations. The disadvantage of this method is that we can reciprocate the similar blocks from two different authenticated images which are protected by the same key. To solve this problem, author uses watermarking based on the content of the image. This can be done by the robust bit extraction algorithm proposed by Fridrich [50]. In this algorithm, robust bits are extracted from each block of the image. This robustness ensures that all comparable blocks deliver the same bits. This effect is helpful in producing pseudorandom watermark successions used in the present watermarking scheme without access to the hash and the original image. The security feature is provided by the generation of a secret key. This secret key depends on the bit extraction approach. Self-embedding watermark is conducted to solve copyright problem Kim et al. proposed self-embedding scheme to detect the tampered area using AMBTC and OPAP techniuqes. Here checksum is used to authenticate, AMBTC is used to recovery the bit of tampered area [77].

b) Block based

To determine the precise tampering region, block-based methods are used. Where the image is divided into 64 * 64 pixel blocks and the watermark is inserted for each block. To check the image is authentic or not, the user tests the absence or presence of watermark in all blocks. The tested image is said to be authentic if the presence of a watermark in each block is highly likely. The VW2D method [163, 164] is based on a block-based watermark that is employed for authenticating the image. Here, the authors have proposed two invisible watermarking methods, namely the addition of an n-binary sequence and filtering. In addition to the n-binary sequence, each image block is assigned a pseudo random number between 0 to 1 and − 1 to 1. The advantage of this type of watermarking is that if the user knows the watermark, then only he/she can extract the image or else an unappreciated image is obtained. The results in the article show that minor changes in the images can be detected. The proposed method is resistant to both nonlinear and linear fitters, as well as JPEG compression. Fridrich [49] presented a similar technique but a novel method for protecting the content of the image using the self-embedding method. The cropped, resized or tampered image has been recovered using the DCT with a window size of 8 * 8 pixels. The result shows that, after the image is recovered, the quality of the recovered image is reduced to 50 % when compared to the quality of the original image. To intercept the unauthorized removal or purposeful distortion of the watermark, the author prescribes making the watermark reliant on the image in which it is implanted. The binary signature utilized is compared to a pseudo-irregular sign created by a secret key. The content of the block and the number of blocks are represented by the M-tuple of bits. Utilizing the O Ruanaidh spread range technique [121], all blocks are watermarked. The author asserts that the watermark is robust to any changes with reference to noise distortions, JPEG compression, cropping and brightness, etc. These watermark properties empower us to recognize the difference between manipulated images and non-manipulated images. For effective search of tampered region, sliding window is introduced [28] where the image is divided into number of non-overlapping blocks; each block is arranged in low level

hierarchy. This method helps to improve security against tampering, using the hierarchical structure concept and the vector quantization counterfeiting method. To locate the signature tampering in the image, the image is divided into smaller blocks. When the search area is divided into smaller blocks, searching becomes easier, and the sliding window technique does not lose any pixel locations, increasing the accuracy of image authentication. To accurately detect the tampered area Dadkhah et al. proposed a block based technique where image is divided into number of blocks and 16 bit key is generated from each block of the image. Accurate tamper detection with less computational complexity is the advantage of proposed technique [38]. To authenticate the content of industrial image author presents DWFCAT framework. Where image is divided into non-overlapping blocks and embed the information into each blocks to obtain the watermarked image. This framework provides robustness against different attacks and helps to detect the tampered area exactly [72].

2.1.2 Semi-fragile watermarking

A semi-fragile watermark is a type of authentication watermark. This method is less sensitive to JPEG compression and more robust when compared to a fragile watermark. The main intent of this method is to segregate between harmful manipulations, like the addition or deletion of important features of the image, and any other processes which should not change the semantics of the image. The main reason for choosing this method is due to the fact that the image is transformed and stored in a constricted form. The loss due to the constriction process doesn't alter the integrity of the image and also its interpretation. To overcome the drawback of block based watermark, semi-fragile method was introduced [131]. Where in block based watermark, the payload of watermark is high but in semi-fragile watermark it is low and helps to detect and locate malicious tampering in images. The wavelet coefficients selected for embedding are randomly permuted with a secret key, achieving high security and tampering detection with less prior knowledge of the watermarks.

The semi-fragile watermarking for integrity verification uses a quantization technique. This quantization technique also helps to locate the tampered region accurately. A telltale tamper-proofing technique [79] is employed to detect whether the image is tamped or not by collecting the spatial and frequency domain features from the image. The user can add a watermark to the image using DWT quantization by "user specific keys". The simulation results show that the method has efficiently localized the region of tampering. In normal quantization-based techniques, some of the unauthorized modifications cannot be observed. A sequential quantization strategy (SQS) [36] is suggested to render the host value modulation based on a certain amount of the previous ones. Any change in the host value may alter the data embedding. This SQS is implemented with the reversible watermark technique, so that a new approach is generated and reversibility is achieved. This SQS detects tampered regions, which aids in the integrity verification.

A new approach is proposed by Bolourian et al. [24] using semi-fragile watermark based on LWT and FNN for authenticating the image and localizing the tampered area. Threshold value is used to distinguish between blocks and LWT transform is chosen over other transforms because of low computational complexity and increases the robustness. FNN is used because of its powerful feature extraction which improves the true positive rate [24]. To improve the robustness of the image, Content dependent semi-fragile watermarking technique is proposed. Where the watermark is generated based on the image content. To embed the watermark image

is divided into non-overlapping blocks. Secret key is embedded into divided blocks and error map is used to detect the tampered area [63].

2.1.3 Robust watermarking

Robust watermarking is also a type of watermarking where it can resist non-malicious operations. Robust watermark can resist against scaling, filtering, compression and geometric distortions etc. A novel method is proposed by Weng et al. where it uses robust watermark to embed into content of the image [162]. This scheme includes hash algorithm and watermark technique to authenticate the image. Firstly, the image is resized into 512*512 pixels and divided into 64*64 blocks from each block hash is generated by applying 2-D DCT and DWT. Then extract the DCT coefficient using zigzag scan and concatenate the bits extracted. QIM watermark technique is used to embed the watermark into corresponding blocks. This QIM technique is used to protect the image quality. This method performs well without compromising image quality. Later double stage contourlet- based watermarking is introduced [134]. This method includes 2-stages embedding where image is divided into non-overlapping blocks. In first stage high frequency signature is embedded using local contourlet transform, whereas in second stage at signature is embedded in low frequency using global contourlet transform. This method achieves good robustness compare to other watermarking techniques by embedding watermark in local and global level.

Tamper detection based on threshold value is proposed by Laouamer et al. using robust watermarking [81], where image is divided into non-overlapping blocks and from each block pixel is chosen to embed watermark in spatial domain using weber law. This scheme aims to give robustness with tamper detection. Here tamper detection is done by extracting the attacked watermark. This technique is robust to geometric and non-geometric attacks. Spatial domain relatively has less robustness and low capacity to hide information. To increase the robustness author Byun et al. introduces a novel method using DCT which gives high robustness with low computational complexity [26]. To embed the watermark into the host image firstly select the embed block and apply DCT for specific location. Calculate the variation from selected block. Then embed the watermark into image. This method outperforms for two publicly available datasets [CVG-UGR and USC-SIPI].

2.2 Digital signature

The concept of the digital signature was introduced to provide an image authentication service. The most straightforward approach for validating the content of an image is to use a cryptographic hash function. Image features or pixel values are fed into the hash function. The generated hash code is then encrypted using an asymmetric cipher. The resultant signature is commonly stored as metadata for future verification. This signature yields a decision with respect to image authenticity as shown in Fig.2. Digital signature method can be broadly categorized into (i) Cryptographic signature (ii)Robust signature (iii) Perceptual hash (iv) Forensic Hashing for Information Assurance[FASHION].

2.2.1 Cryptographic signature

Many signature techniques have been developed to ensure the authorization of the image's owner [98]. When images are used illegally, these technologies may allow the

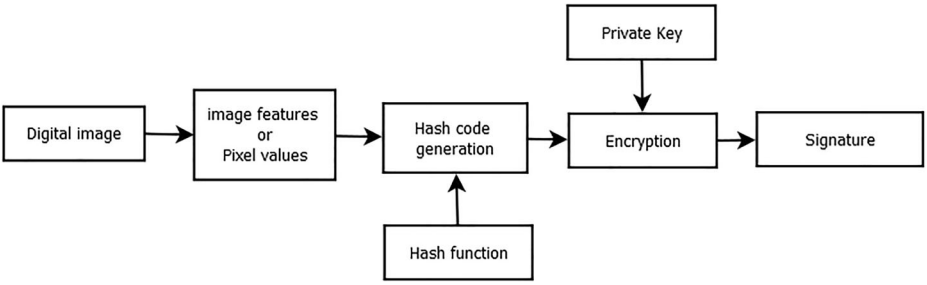

Fig. 2 Basic Digital Signature generation procedure

image owner to claim copyright. Images are typically compressed before transmission in this sender-receiver protocol. It provides a public-key based fault-tolerant and compression-tolerant digital signature. It also recovers the image when the modifications happen. All signature-based methods intended to reduce the length of the hash it produces as much as possible. To shorten the length, the Karhunen–Loeve transformation [156] is employed. It will first use DCT and DWT to extract the features. Then these features are encrypted using a sub-key generated by the Ripemd-160 hash function. This provides a tool for tamper localization and the hash length is independent of the image size. It can also increase their detection efficiency by using the image thumbnail [170]. The majority of contemporary models are designed to detect a certain type of tampering. Hence, a quaternion-based image is proposed in the article and it has defined almost all types of distortions. This method is a combination of structural and Color features. These features are extracted to propose the quaternion Fourier transform (QFT). QFT helps to calculate the feature hash using a hash function. It is easy to locate the tampered region. The goal is to provide a quick and efficient way of authenticating an image, which includes being resistant to numerous content-preserving attacks.

2.2.2 Robust signature

A robust signature technique is to generate unpredictable randomized hash signatures for authorized users [151] to secure the hash value, the encryption of chaotic sequences of the block DCT coefficients is used. Principal Component Analysis (PCA) is used to encrypt the DCT co-efficient. To generate a hash value, the Eigen matrix is used. The length of the hash value is 72 bytes. Experiment on a large database shown that the technique is efficient.

Based on a randomized hash method, any modification in the image can be located. To overcome the weakness in the feature extraction of block-based hashing, a new wavelet-based hashing is proposed in [6]. The KI transformation is applied so that it uses a secret key which directly modulates the image pixels. Although this transformation gives good results for JPEG compression, it's still sensitive to content preservation attacks. To achieve robustness against content preserving modifications and detection of malicious operations, the discrete wavelet transform [DWT] and the radon transform are employed [58]. As a first step, image features are extracted by using DWT and the radon transform. Then, to map those features into a binary sequence, probabilistic quantization is used. So, these results are robust against filtering, scaling and can be able to detect malicious tampering. An image hash should be robust for legal operations and not prone to unintended abuse. A method proposed in [137] for authenticating the image with small

hash and it also locates the position of tampered area. Hash-based image authentication [7] aims to solve core problems such as tamper detection, security and robustness. All other methods use a secret key to generate a sub-key, but here, a secret key is used to modulate the image pixels randomly to create a transformed feature space. The key-dependent transform is used to calculate the image hash. A 4-bit quantization scheme is used to reduce the hash length. Extracting features from an image is a time-consuming task, and extracting robust features is even more difficult. [183] employs Zernike moments, which are effective at extracting robust features from images. These Zernike moments are based on rotation invariance of magnitude. Similarity between hashes is calculated by hashing distance. The authentication of non-aligned images is critical for reducing fraud [7, 183]. Battiato et al. [19] proposed an alignment method where image hashes are generated by using a bag of features. Firstly, the image is divided into blocks and then, for each block, HOG features are extracted and feature codes are generated. From each pixel of each block, the magnitude and orientation are computed. The magnitude indicates the presence of edges. To obtain the hash for each block, histograms are normalized. In order to perform tampering localization, the receiver has to take out all the geometric transformations and align them with the senders' image. This alignment should be done in a semi-blind way. A good alignment gives accurate tampering detection. It can be able to detect minute tampering with the localization of the tampered region. It is robust against JPEG compression and low-pass or high-pass filtering subjected to the transmitted or stored image.

2.2.3 Perceptual hash

To solve the problem of content authentication, perceptual image hash techniques are widely used because they are sensitive to tampering and luminance changes. The scheme in [159] includes hash generation, tamper detection and localization. Watson's visual model is used to extract visually sensitive features from the image. Then key point based and image block based techniques are employed to generate the perceptual hash. To generate the final hash code, Huffman coding is used. Finally, the distance metric between the original and transmitted images is calculated to detect whether the image is tampered with or not and to locate the forged regions. To tolerate high geometric distortions, an image alignment process is used [161]. This scheme presents a perceptual image hash based on image alignment and tamper detection with localization. Firstly, image features are extracted using global, local, Zernike and combined with DCT co-efficients. These hybrid features are extremely sensitive to even minor changes. Encryption helps to hide the hash code and image content. Randomization is used to increase the randomness so that collisions are reduced when generating a secret key. A hash code is generated by reducing the hash length using quantization and compression. Image hashes and local features help to estimate the changes that have occurred in the image. Multi-media hazard is an effective solution to provide proof of legitimacy and possibly identify the manipulation that has been performed [150]. This algorithm is based on compressive sensing principles that resolve both authentication and tampering detection problems. When the image's brightness, affine wrapping, and contrast are authenticated using distributed source coding. This distributed source coding gives the desired robustness against unauthorized manipulations. The proposed algorithm is robust to moderate content-preserving transformations, including cuts, scaling and rotation.

2.2.4 FASHION [forensic hashing for information assurance]

Nowadays, transmitting information via the internet is common, and information may be used as evidence in some cases. Securing information from unauthorized users is very important. It is very important to create trust for online transactions. In 1924, however, a new framework called FASHION, which stands for forensic hashing for information assurance, was proposed. A hash is a short hash which is attached to the image before transmission and it acts as a processing history for the received image. Visual words are used to generate a hash, which keeps the hash length short and the alignment component is used in a forensic hash to estimate geometric transformations. They generate SIFT points and cluster those points to create a final hash that is utilized to estimate geometric transformations. Lu et al. [101] proposed new methods for carrying out multimedia forensics using compact side information to reconstruct a document's processing history. The suggested forensic hash is generated based on the Radon transform and scale space theory. Using 30–40 bytes of hash value, it can accurately detect the tampered region. This is robust against multiple operations. The primary goal of this project was to generate a forensic hash for compact data. As stated in [101, 102], aligning the image is a common step to avoid tampering. The scheme proposed in [20] uses a bag of visual words [BOVW] for representation. It helps to reduce the feature descriptors. During the training phase, the codebook is generated using the SIFT points. This codebook is generated once and will be used for communication between the sender and the receiver. First, the sender extracts SIFT points and arranges them in descending order before selecting and labeling the top SIFT point with a codebook. Hence, the final hash is generated for each SIFT point, and the ID, the dominant feature and the scale are all considered. Then this final hash is attached to the image and sent to the receiver. After sending information over the transmission medium, the same procedure has to be followed by the receiver to get the original image. If the hash value varies, the image is considered to have been tampered with by an unauthorized user.

To generate a robust signature, global and local features are extracted from the image. The local features refer to distinct structures found in images, like blobs, edges, corners, regions and small patches etc. From those features, a hash code is generated that helps to authenticate the image. This local feature extraction mechanism is resistant to geometric transformations. In [105] a novel shape-contexts-based image hashing approach has been proposed that utilizes robust local feature points. Firstly, the SIFT points are used to generate the feature points, followed by the Harris criterion to select the most stable points. Secondly, the image's local information is characterized using those local features, and shape contexts are introduced to generate the image hash. This shape context based hash helps to detect and locate the tampered region.

A new image hash algorithm [146] is proposed to enhance the tamper detection capability by employing one of the most well-known local feature descriptors. Image intensity transformations using a random number generator, Successive Mean Quantization Transform (SMQT), and bit-level permutation are utilized to obtain a secure and robust hash value. The main problem addressed is the robust tampering detection of the image received in the transmission under various content-preserving attacks [132, 171]. The multi-scale image hashing method is proposed based on the features generated by location context information. The generated hash is attached to the image before transmission and analysed at the destination to filter out the geometric transformations that have occurred in the received image. If the received image is tampered with, then the tampering regions will be localized through the multi-scale hash component. Experimental results on different images with tampering of

arbitrary size and location demonstrate that these image authentication and tampering localization schemes are superior to the state-of-the-art methods under various attacks.

Global features refer to the whole image, representing texture, color and shape information, etc. Here, the Local Binary Pattern (LBP) is used because of its high texture description ability and its low computational complexity. Feature extraction is the main step in all perceptual image hashing schemes [39]. This scheme proposes a novel approach where it uses CRLBP for feature extraction from each block. The product of the weight vector and CRLBP gives a final hash. To check whether the image is authentic or not, a comparison of the hash values pertaining to the original image and the transmitted image is done. If hash values are same then it is authentic otherwise the image is fake and it leads to tampering detection. Tampering localization can be done by measuring the hash distance. To precisely locate the tampered region, LBP presents a modified technique that is Noise Resistant Local Binary Pattern (NRLBP) [1]. This NRLBP, combined with SVD, is used to generate more robust features which are more sensitive to small changes. This approach identifies tampering as small as 3% of the image size.

For content authentication, a novel forensic signature method has been proposed. It is scalable and compact by selecting the proper robust features from the original image. A forensic signature is generated based on statistics of feature points [157]. The original image feature points are first extracted using corner detection methods such as traditional Harris corner detection and adaptive corner detection. These methods evaluate the features which are generated by different techniques. Although Harris detection is advantageous, it is still susceptible to noise. They pre-process the image using DWT, to avoid noise. Then, extract features from the reconstructed image using a Harris operator. Then a feature point in the neighbourhood is selected to cover the whole image. Using a feature point neighbourhood, a forensic signature is generated by extracting the statistical features. The Huffman code is used to compress the signature. Then, the signature is encrypted using a public cryptosystem, to enhance security. The Fisher criterion is used to generate the threshold value. Based on that threshold value, it is checked that the image is authentic or tampered with. To detect and locate the tampered region, this threshold value is used. This scheme is robust against geometric distortions, noise filtering and JPEG compression etc.

The Fourier fractional transform (FRFT) [138] is proposed to improve sensitivity against illegal manipulations. This method includes three steps: pre-processing, feature extraction and post-processing. Pre-processing always includes rescaling, a low-pass filter and adjusting the component, and so on. The most important step in creating an image hash is feature extraction. To achieve more security, compactness and robustness, the feature extraction step plays a very important role. FRFT is applied to every block of the image; from each block, a secret key is generated. Summing up all the co-efficient to generate the hash code. Finally, a post-processing step is applied to reduce the length of the image hash. Experimental results show that the proposed scheme is more accurate in detecting tampering with localization and more robust against rotation, mid-filtering and JPEG compression. This scheme is more suitable for CBIR applications.

Image hashing is a technique which generates a simple, fixed-length hash code to represent the content of the image. Therefore, it is used to authenticate the image. These hashes are constructed by cryptographic algorithms. Usually, these algorithms are sensitive to small changes in the input data. The image hash should be short and sensitive to illegal manipulations and robust to affine transformations. Each hash code must be different for different images. As a result, unauthorized individuals are unable to breach the security. To achieve all these requirements simultaneously is a challenging task. So the method of combining local and

global features has more advantages for meeting this entire requirements. It includes the following steps: pre-processing, global feature extraction, local feature extraction, shape feature extraction and hash construction. The aim of pre-processing is to eliminate the noise so that the generated image has the same computation complexity and a fixed size. After pre-processing, extract the global features from the image, which include Zernike moments [185] of the chrominance/luminance and then local features are extracted from the salient regions. The salient region is nothing but the visual attention and texture which are part of human visual perception. The local features are skewness, kurtosis, contrast and coarseness. This feature extraction gives an intermediate hash. The final hash is generated using a pseudo-random technique. The hash of a test image is compared with that of a reference image. When the hash distance exceeds a certain threshold, the received image is deemed a forgery. By decomposing the hashes, the type of image authentication and location of forged areas can be determined. This method is robust against content-preserving operations; the hash is sensitive to malicious tampering. Searching for the correct salient region is difficult, so Gabor wavelets [16] are proposed to identify the salient regions. The Gabor wavelet extracts global features from a global perspective. Then, local features are extracted from the image histogram and a final hash is generated from these features. The hash of a test image is compared with that of a reference image. When the hash distance exceeds a certain threshold, the received image is deemed a forgery. Being robust against content-preserving operations, the hash is sensitive to malicious tampering and suitable for image authentication.

2.3 Hybrid technique

A hybrid method is a combination of two or more distinct techniques. As illustrated in Fig.3, a hash value is first generated from the original image, and then the hash value or image features obtained from the original image is/are embedded as a watermark and sent to the receiver. If there is any variation in the hash value at the receiver end, it is considered to be tampered with. Chen et al. [32] uses a combination of digital watermarking and digital signature techniques to prove the authenticity and the integrity of an image. Digital watermarking embeds the watermark in the image, whereas a digital signature generates a separate signature file and transmits it to the receiver with the image. To avoid this extra process and to enhance accuracy, the author combined the digital watermark with the signature. The recommended approach generates the signature from the image and embeds it as a watermark in the original image. As a result, the additional step of sending the signature file together with the image is not required. This method provides not only integrity verification but also accurate tampering localization.

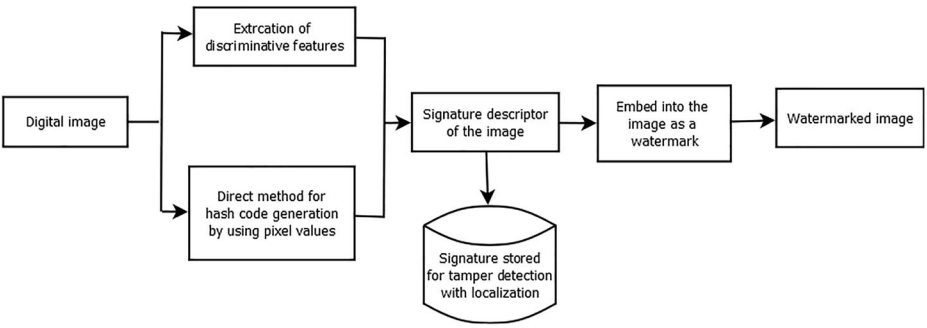

Fig. 3 Block diagram of hybrid technique

[180] Introduces a binary watermark consisting of image texture information which is generated and inserted into the least significant bit (LSB) of the original host image. To improve the security, the Arnold transform is employed twice in embedding process. Here pixel based watermarking is presented for authenticating the image. To detect the tampered region, the comparison between original image and tampered image is taken. Change in the single pixel leads to tampered image. It can detect and locate the tampered region effectively for various malicious attacks. The algorithm has high security and perceptual invisibility.

Digital watermarking includes visible and invisible watermarking techniques with a cryptographic hash function. A new approach to an invisible watermarking technique is being proposed in [174] which are helpful in image verification. Here, image pixel values are embedded with an invisible watermark. It includes two processes: one is stamping the original image with an invisible watermark and getting a verification key. The second one is based on the verification key extraction of the watermark from the original image. Then the extracted watermarked image is used for verification. If the extracted watermarked image and the original image are the same, then it is said to be authentic. Otherwise, it is a tampered image. Later, Wong et al. [166] presented user-defined watermarking methods, namely visible and invisible watermarking techniques, to authenticate and verify the user. The watermark extraction is successful only when the user provides an accurate key or else the pseudo random image is pushed to the user. The algorithm supports the transmission of two keys: one public and one private. It requires a secret key in the insertion and extraction process. Security resides with the user's key; it's not dependent on the algorithm. For secret key generation, MD5 is used. This scheme may be able to detect and locate the modified region in the image. To improve tampering localization, author Celik [29] presents a new framework that is called lossless authentication watermarking [LAW] for image verification. The main advantage of the LAW framework is that it provides a verification step before the reconstruction step. If the image meets certain criteria for the original image, then there is no need for reconstruction. The LAW framework supports public and private mechanisms. Another significant advantage of this framework is the ability to pinpoint the affected area. The LAW framework divides the image into blocks, and for each block, a key is embedded; if the key changes, it identifies that position as having been tampered with.

To check the robustness against image compressions Lin and Chang et al. [86] proposed an algorithm that is robust to JPEG lossy compression to a pre-determined quality factor and resists malicious attacks. They have given importance to two constant properties of DCT co-efficients with JPEG compression. The first property tells us that if we modify the DCT co-efficient by multiplying with an integer (Quantization step), later, this can be recovered exactly after the JPEG compression. The second property says that there will be a relationship between the two similar co-efficients that are before and after the compression of the JPEG. In JPEG compression, all DCT co-efficient matrices are divided by a similar quantization table. The relationship between the DCT co-efficients, which is the same co-ordinate region that belongs to two blocks, does not alter after the process of quantization. The main reason for quantization is that complex inequalities may change into simple equalities.

The properties proposed by Lin and Chang can be used to include the signature of the first property and the second property in order to obtain the authentication bits. The steps of embedding and authentication are as follows. To begin, the original image is divided into 8 * 8 pixel blocks using the secret mapping function, which results in block pairs. For each of the block pairs, generate the binary signature. A binary signature is then halfway embedded into each of the block pairs. For example, if the length of the signature is 8 bits, then 4

authentication bits have to be embedded for each of the block pairs. The embedding process consists of establishing an equal relationship between the LSB of the present DCT coefficients and the signature bits. The authentication process involves extracting bits from the watermarked area of the image and observing whether or not the DCT co-efficient is related to the predicting criteria. If this image is related, then it is considered to be an authentic image. If it is not matched, then either the block or both the blocks are considered to be manipulated. Another important improvement is that it can precisely locate the altered blocks in the image.

In order to see whether the original image is altered or not, it is needed to compare the original image features with the watermarked image. If the image features are similar, then it is concluded as an original image or else it is concluded as a tampered image. The image alterations can be detected by using the image features. These features are based on the type of image, like medical, military, satellite, etc. The images which are under lossy compression and detached due to harmful manipulations are considered as invariant properties for selection of features. These features can also be used to locate the image regions which are tampered with. In order to provide authentication services for images, many features, like luminance differences, edges, colours or the mixture of these features, are used. Many limitations are included in this method, which are in the form of the robustness and the capacity of the signature storage. Robustness is needed to permit lossless extraction of watermarks. There is a direct relation between the amount of information incorporated into the image and the efficiency of image tamper detection. It is always needed to see a better compromise in the signature size to achieve good results in image tamper detection and robustness. Anyhow, with watermark insertion, there is an issue of image modification. Small changes can cause variations in the image features. Then the modified image and the original image are not identical. This could result in false positive detection. The importance of this issue depends on the features considered. To rectify this problem, an alternate watermarking algorithm has been implemented. This method is to sign an image by taking out features from the newly modified images. Then the original image should be watermarked using the new features which are computed. This iterative cycle empowers coincide the features of hidden image with the features of protected image. But this feature selection doesn't have any secret keys. In order to own the secret key at the level of feature extraction, the author Chang et al. proposed a new framework, where features of the image are obtained from the hash algorithm [30], so that only that person can have the valid secret key that proves the ownership of that image. Without a secret key, it is impossible for others to test the ownership. This framework helps to detect and locate the tampered region accurately by comparing the original image with the embedded image. The difference between the pixel values leads to locating the tampered region. Reversible image authentication [148] has drawn great attention to its ability to recover the original image from the watermarked image. The paper describes a block-based reversible watermarking scheme for image authentication based on histogram modification of the differences between adjacent coefficients of DWT. In [12] Anbu et al. explained detailed survey on different image forgery detection methods and concluded that convolution layer gives good result of 99.10% accuracy.

Table 1 provides a comprehensive comparison of active image authentication techniques which enable tamper detection with localization. This table gives a gist of techniques for active authentication, domain on which authentication was done, viz. frequency spatial, input type like pixels or block of pixels, possibility of recovery of the altered/manipulated portion after localization and whether the technique is sensitive to only malicious operation or both malicious and non-malicious operations.

Table 1 A comprehensive comparison of active image authentication techniques

Paper Reference No.	Active Authentication Technique	Basic domain used for image authentication	Pixel/ block based	Image Recoverability	Sensitive to operations
[47]	Hybrid	Frequency	Block	No	Malicious
[173]	Watermarking	Spatial	Pixel	No	Non-malicious
[30]	Hybrid	Frequency	Block	Yes	Non-malicious
[164]	Watermarking	Spatial	Block	No	Non-malicious
[121]	Watermarking	Spatial	Block	No	Malicious
[28]	Watermarking	Frequency	Block	No	Non-malicious
[36]	Hybrid	Spatial and frequency	Pixel	No	Malicious
[165]	Watermarking	Spatial	Block	No	Non-malicious
[168]	Watermarking	Frequency	Block	No	Non-malicious
[124]	Watermarking	Spatial and frequency	Block	No	Non-malicious
[88]	Watermarking	Frequency	Block	No	Non-malicious
[163]	Hybrid	Frequency	Pixel	No	Non-malicious
[50]	Hybrid	Spatial	Block	No	Non-malicious
[79]	Hybrid	Frequency	Pixel	No	Malicious
[86]	Hybrid	Spatial	Pixel	No	Malicious
[174]	Hybrid	Spatial	Block	No	Malicious
[166]	Hybrid	Spatial	Pixel	No	Non-malicious
[29]	Hybrid	Spatial	Block	No	Non-malicious
[131]	Hybrid	Spatial	Block	Yes	Non-malicious
[98]	Signature	Spatial	Block	Yes	Non-malicious
[156]	Signature	Frequency	Block	No	Non-malicious
[170]	Signature	Frequency	Block	No	Malicious
[151]	Signature	Frequency	Block	No	Malicious
[6]	Signature	Frequency	Block	No	Non-malicious
[58]	Signature	Frequency	Block	No	Malicious
[137]	Signature	Frequency	Block	No	Non-malicious
[7]	Signature	Frequency	Pixel	No	Non-malicious
[183]	Signature	Spatial	Block	No	Non-malicious
[19]	Signature	Spatial	Block	No	Non-malicious
[159]	Signature	Frequency	Block	No	Malicious
[161]	Signature	Frequency	Block	No	Malicious
[150]	Signature	Frequency	Block	No	Malicious
[101]	Signature	Spatial	Block	No	Non-malicious
[102]	Signature	Frequency	Block	No	Non-malicious
[20]	Signature	Frequency	Block	No	Malicious
[105]	Signature	Spatial	Pixel	No	Non-malicious
[41]	Hybrid	Frequency	Pixel	Yes	Malicious
[146]	Signature	Frequency	Block	No	Non-malicious
[171]	Signature	Frequency	Pixel	No	Malicious
[132]	Signature	Spatial	Pixel	No	Malicious
[39]	Signature	Spatial	Block	No	Non-malicious
[1]	Signature	Spatial	Block	No	Non-malicious
[157]	Signature	Frequency	Block	No	Malicious
[138]	Signature	Spatial	Block	No	Non-malicious
[185]	Signature	Spatial	Block	No	Malicious
[16]	Signature	Spatial	Block	No	Malicious
[32]	Signature	Spatial	Pixel	No	Malicious

3 Passive approaches

The passive approach to image authentication relies on inherent natural characteristics of the image rather than sender-specific information such as a signature or watermark. In the absence

of explicit authentication, the passive approach is the alternative solution for authenticating the image. Most digital image forensic techniques employ passive techniques to detect evidence of content manipulation in the absence of watermark or signature. It is believed that though digital fake images do not leave any visual clues that have been distorted; they still disrupt the underlying statistical property of the content of a natural image [41].

Passive approaches for image tampering detection can be broadly classified into five type's namely pixel-based, format-based, camera-based, physical environment-based and geometry-based detection [47]. The following section discusses state-of-the-art techniques in each of the tampering detection type which also facilitates localization of the tampered region.

3.1 Pixel based technique

Pixel-based detection is usually dependent on the pixel values of the image. These techniques are used to identify statistical anomalies present at the pixel level, due to manipulation which include three types, namely, copy-move, resampling and splicing techniques. The copy-move method is the common method used in image forgery in which a part of the image is copied and pasted somewhere on the same image. Resampling is simply resizing, stretching, or rotating a portion of the image onto a different testing grid. Splicing is a type of image forgery where at least two images are composed to form a single image. Perhaps, it is very hard to detect the spliced region by visual inspection, when two images are composed to form a single image.

Many publications have suggested various methods for detecting pixel-based forgeries [11, 18, 27, 33, 37, 40, 45, 51, 52, 54, 57, 61, 62, 64, 65, 71, 73, 80, 83, 85, 87, 89, 90, 93–95, 104, 107, 112, 114–118, 123, 127, 133, 136, 143–145, 149, 155, 167, 176–178, 182, 184]. Initially, the original image is divided into non-overlapping blocks and then features are extracted from each block using PCA, DCT, DWT, FMT, SVD, SIFT or SURF. To detect the tampered regions, feature selection and lexicographic sorting will be performed. Fridrich et al. proposed a DCT based method for detecting the copy-move region. However, it is ineffective for noisy images [127]. The PCA approach was utilized in [104] to automatically determine the precise copy-move region, although the time complexity was significant. The similarity matching approach is utilized to minimize time complexity, and it also works in noisy conditions [104]. Zhang et al. present a unique strategy that works for both noisy and compressed images and accurately detects the copy-move region using the DWT method [176]. Then, in comparison to other algorithms, DWT and SVD are coupled to minimize time complexity [83]. Later, SIFT features are utilized to identify not just the copy-move area, but also the false region [64]. As a result, SVD can reliably detect the copy-move region, but it will not perform well in images that are heavily noised or compressed [73]. PCA has been enhanced to operate with such substantial noise- and compressed images, allowing it to precisely locate the copy-move region [89].Splicing methods are also concerned with detecting the spliced region. Therefore, DW-VAM [133] was presented by z.qu et al. to identify the spliced region. The DCT-SURF technique is used to detect both splicing and copy-move. It aids in the precise detection of manipulated regions [87]. Combining DCT and DWT may be used to detect highly compressed image forgery; it will accurately detect the fake region [54]. DCT is utilized to identify the forged region in a blurred, noisy, and compressed image and helps to detect the exact location of the forged region [27, 65]. Tian-Tsang et al. [116] developed a novel splicing detection technique that uses bicoherence characteristics to identify the splicing. Chennamma et al. [35] exploited lens radial distortion to develop a novel approach for detecting splicing. If

there is any inconsistency in the degree of lens radial distortion across the image, then image manipulation can be detected using linear-based calibration. This method works well for real images too. Splicing detection using a Markov method was proposed by Zhongwei He et al. [62]. Firstly, extract Markov features from the original image in both the DCT and DWT domains. Then, the Markov features extracted in the DCT domain are more extensive since they include not only intra block, but also inter block correlation. After generating all related features, they utilized SVM-RFE to choose features to reduce dimensionality and produce the final vector. Finally, SVM is employed as a classifier to discriminate between legitimate and spliced images. Mohammad Manzurul Islam et al. [66] proposed the LBP texture operator for splicing detection in color images. The image is divided into non-overlapping blocks, and the DCT is used to identify changes in the image caused by forgery, while the LBP is used to improve tamper detection. The resultant LBP image is again divided into non-overlapping blocks and finally arranged as a feature vector.

Generally, image forgeries are created by copying and pasting some portion of an image onto another (image compositing). While doing so, resizing of the image is required to match the original image grid. This process is known as resampling. This resampling process introduces a new correlation into the image. This aids in the detection of the modified portion. [129] proposes an EM algorithm for estimating the probability map and including the correlation between each pixel and its neighbors, which aids in the detection of tampered regions. The primary disadvantage of this method is that it can only be used with uncompressed or minimally compressed JPEG, GIF, and TIFF images. To overcome this drawback, Prasad et al. [130] proposed a DCT and DWT based approach in which DCT is applied to the entire image to retain the high frequency coefficients. This assists in detecting tampered regions. Wavelets can be used to detect the tampered region in which the image is separated into three details: horizontal, diagonal, and vertical, instead of maintaining high frequency coefficients. After extracting these three details, the remaining coefficients are eliminated. This method is more resistant to resizing and compression.

Babak Mahdian et al. [108] proposed a novel method based on a radon transform and derivative operation, which is an efficient, blind, and automated technique for tracing interpolation and resampling. This approach can be utilized to detect forgeries when two or more images are spliced together to create a quality forgery, which also includes geometric transforms like scaling, skewness, rotation and resizing. Interpolation and sampling steps are required for detecting these geometric transformations. By having the resampling/interpolation detector, the manipulated region may be easily detected. This method is more robust against JPEG compression and resampling, but the time complexity is higher. In order to reduce time, Matthias Kirchner et al. [78] proposed fast and reliable resampling by fixed linear predictor residue. Here, transformation parameters influence the residue that helps to detect the exact resampling region in the image. When compared to other techniques, the experimental results show that it is a reliable method.

In order to know the state-of-the-art methodologies for image forgery detection, a few review papers can be found in the literature which highlights pros and cons of different forgery methods based on passive approach [4, 14, 139, 154]. In recent years deep learning technique has been widely used to detect and localize the forged region. Rao et al. uses CNN technique it is specially made for splicing and copy-move detection application [135]. Where CNN model is pre-trained on labeled samples it serves as local patch descriptor. Dense features are extracted from test image using this pre-trained CNN and for SVM classification feature fusion method are incorporated. To detect the tampered region accurately Mohammed et al.

introduce discrete cosine transform [10]. Where, firstly it converts RGB image into gray scale then divide the image into non-overlapping blocks. For each block 2D DCT coefficient is computed and using zig-zag scanning, feature vector is applied on 2D coefficient. Then sort the blocks using lexicographic sort next each block is matched to detect the duplicate blocks using Euclidean distance. Not only block matching is sufficient to detect the forged area to localize the forged area author uses shift vector to make the forgery decision.

Resampling features and deep learning are used to detect the manipulation in image. Firstly tampered regions are localized using Random Walker segmentation method. Then resampling features are computed through LSTM for localization. Both efficiently result in detecting copy-move, resampling and splicing detection [25]. To boost overall forgery detection process author Mohammed et al. proposed resampling detection framework where copy-move algorithm is used as pre-filtering step to boost the forgery detection [110]. Not only copy-move is important to detect, digital splicing also plays a vital role in manipulating the image. In [92] author proposed a deep fusion network to locate the splicing forgery where set of convolution layers are combined to form a fusion network. This fusion network achieves effective result in numerous situations. A new approach for tampering detection has been introduced by Vega et al. [13] where the combination of DCT, DWT and LBP is used to detect the forged area. Firstly image is converted into YCbCr then DWT is applied. The converted image is then divided into number of blocks and for each block histogram is applied then LBP features are extracted. From each LBP block DCT coefficients are extracted then vectors are concatenated and SVM is applied. This method is applied to CASIA v1.0 and CASIA v2.0 dataset to achieve good result. A deep learning based convolutional neural network was introduced by muzaffer [113] where input image fed into convolutional neural network to extract features and then feature matching is applied to know about forgery existence. To eliminate the false match post processing is applied and this helps to localize the forged area. Copy-move, splicing and removal manipulations are detected using skip connection framework [109]. This framework consists of three layers LSTM, encoder-decoder and skip connection. In first layer LSTM it learns the transition between tampered and non-tampered blocks. Encoder-decoder consists of convolution to learn about spatial information. Last layer takes both LSTM output value and spatial details for tamper detection.

Convolutional neural network plays a vital role in detecting tampered area and provide promising results compare to other techniques. The neural network is trained in such a way to detect the tampered area. Firstly images is fed into CNN then features are extracted and then features are compared to detect the forged area. GAN [Generative adversarial networks] are used to detect the forged region [2]. In [82] the authors present a CNN to detect the deep Inpainting in digital images. Where features are extracted pixel-wise localization is done for Inpainting images and this CNN works well for large number of data meanwhile provide tampering localization [59, 96]. To make it more efficient, an updated version of CNN is introduced that is mask R-CNN. This mask R-CNN constitutes 3 stages where first one is features extraction it extract the prominent features from the image. Second one is region proposal and at last prediction which helps to detect the tampered region [8, 160]. To detect the multiple copy-move tampering, Bilal et al. proposed a technique to detect and localize the tampered area. Firstly image is converted into gray scale and DWT is applied to converted image. Then using SURF and BRISK, features are extracted next step is to match the features after that clustering will be done to know about the false match using DBSCAN and RANSAC. Finally post processing is applied to localize the forged area [23]. Bifurcating between tampered and non-tampered image Azrak et al. proposed a trigonometric transform to

localize the forged area and deep learning is applied to distinguish between forged and non-forged image [9].

Machine learning algorithms also give good results similar to deep learning techniques. To detect copy-move and splicing forgery author Lourembam et al. proposed a method using machine learning, Where RGB image converted to gray scale and divide the image into number of blocks. Extract the features from each block using SIFT and SURF and calculate the feature vector value from each key-point. Compare feature vector with each block and detect the forged area based on certain threshold value [43]. Further Doegar et al. proposed a combined technique to enhance the localization. Where, deep learning and machine learning algorithms are combined to give result. For image feature extraction GoogleNet is used and random forest is used to detect the forged area [99]. Recovery of forged area is more challenging. Ying et al. proposed Imuge technique where it combines U-Net encoder with forgery localization network. This techniques not only localizes the forged area but also recovers the original data approximately [175].

Deep fakes are replacing the original face with other face in image or video. Detecting deep fakes is also important where bifurcating between tampered image and non-tampered image helps to make good decision in case of any serious issues. Deep learning with GAN helps to detect the tampered image [56]. Deepfakes survey has been made to know about techniques involved in creation and detection of deep fakes using deep learning [120]. In case of multimedia digital images are more prone to get tampered so providing a tool to avoid deep fakes is important. Luisa Verdoliva gives an idea about different methods, limitation of current tools and future directions about deep fake detection [152].

Table 2 presents a bird view of comparison among pixel-based techniques in terms of approach/method adopted, type of forgery detected, limitations or advantages of each approach along with dataset used for forgery detection.

3.2 Format based technique

One of the most significant image credibility verification approaches is format-based detection. These techniques are most commonly used on images which are in the JPEG format. There are three types, namely, JPEG quantization, Double JPEG and JPEG blocking. JPEG quantization is the process of reducing the number of bits used to compress data to a single quantum number. Most images are subject to multiple JPEG compressions, which mean that once the image is compressed again, it is resaved, resulting in double JPEG compression. JPEG blocking is nothing but when the JPEG compressed image is cropped due to change in inter-block, there will be correlation between the DCT co-efficient. That correlation helps to detect the blocking artifacts [22, 31, 53, 84, 91, 103, 125, 128, 158]. Table 3 presents an overview among format-based techniques in terms of method adopted, type of forgery detected, limitations or advantages of each approach along with dataset used for forgery detection.

3.3 Camera based technique

When an image is captured by a digital camera, the captured scene moves from the camera sensor to the memory. The color filter array, JPEG compression, quantization, white balance, and gamma correction are some of the phases it will undergo while processing an image inside the camera. Different camera artifacts, such as chromatic aberration, color filter array interpolation, camera response, and sensor noise, will be inherent to the image during the processing

Table 2 A comparison among pixel based techniques

Ref.No.	Method	Type of forgery detected	Remarks	Dataset
[51]	DCT	Copy-move	Won't work for noisy images.	–
[127]	PCA	Copy-move	Time complexity is high	–
[104]	Similarity matching	Copy-move	Compare to [127] time complexity is reduced	–
[83]	DWT-SVD	Copy-move	Time complexity is less compare to [104, 127]	–
[176]	DWT	Copy-move	Works well for both noisy and compressed images.	–
[64]	SIFT	Copy-move	Resulted with false positives	–
[73]	SVD	Copy-move	Not able to detect tampered area for noisy and compressed image.	–
[89]	Updated PCA	Copy-move	Good for noisy and compressed images.	–
[87]	DW-VAM	Splicing	Works only for spliced images.	ISCAS images datasets
[91]	Double quantization-DCT	Splicing	Tampered region is detected only in JPEG format	–
[54]	DCT-DWT	Splicing	Won't work for highly compressed image.	–
[133]	DSCT-SURF	Copy-move and splicing	Works good for both spliced and copy-move	Columbia image splicing detection dataset.
[65]	DCT	Copy-move	Works good for blurred and compressed image	–
[27]	DCT	Copy-move	Detects exact tampered region in both blurred and noisy images.	DVMM
[135]	CNN	Copy-move and splicing	For tampering detection CNN has customized design.	DVMM
[10]	DCT	Copy-move	Helps to detect the tampered area accurately using DCT.	CoMoFoD
[25]	LSTM	Copy-move and splicing	Combination of deep learning and resampling features can be done.	Nimble 2016 dataset
[110]	Copy-move detection algorithm	Copy-move and Resampling	Improved the copy-move forgery algorithm.	Nimble
[92]	Deep fusion network	Splicing.	For better results advance convolutional layers can be implemented.	VOC2012 dataset
[13]	DCT, DWT and LBP	Copy-move	Large dataset size for accurate results	CASIA v1.0 and CASIA v2.0
[113]	CNN	Copy-move	False matching is eliminated using post processing	GRIP
[109]	LSTM	Copy-move and splicing	Doesn't detect forgery artifacts but it will localize the forged area.	NIST-16, CASIA 2.0
[23]	DWT, DBSCAN and RANSAC	Copy-move	It can exactly detect the forged area even after rotation, scaling etc.	CoMoFoD, MICC-F220,and MICC-F2000
[9]	CNN, Trigonometric transform	Copy-move	Achieves good robustness in feature matching	MICC-F220
[99]	GoogleNet and Random forest	Copy-move	Achieves good result by combining deep learning and machine learning algorithms	MICC-F220
[43]	SURF, SIFT and SVM	Copy-move and splicing	Low computational load	CASIA
[175]	U-Net	Copy-move	Achieves recovery of forged area	ILSVRC2017 and COCO

Table 3 A comparison among format based techniques

Ref.No.	Method	Type of forgery detected	Remarks	Dataset
[103]	DCT, Neural network	Double JPEG compression.	Estimation of high frequency is not possible	–
[128]	DCT	Double JPEF compression and JPEG quantization.	Robustness of different existing techniques are explained	–
[31]	DCT	Recompression and cropping	Combined spatial and frequency feature to accurately detect the tampered area.	–
[53]	Block-DCT	JPEG compression	Statistical model is proposed	Harrison, UCID
[84]	DCT	Double JPEG compression	Scheme distinguishes between single compressed and double compressed	–
[125]	DCT, SVM	Double JPEG compression	Complete solution for JPEG compression in steganalysis.	–
[91]	DCT	JPEG compression	Fine-grained detection is possible using DCT coefficient	–
[158]	DCT	JPEG quantization	Tampering localization is improved using DCT coefficient	UCID
[22]	DCT	Double JPEG compression	Less computational complexity	Own dataset consist of 100 non compressed TIFF images

steps from capturing to storing in the camera. The discrepancies in these artifacts might subsequently be utilized as proof of tampering [5, 21, 42, 46, 48, 75, 76, 97, 140–142]. Table 4 presents aerial view of comparison among camera-based techniques in terms of approach/method adopted, type of forgery detected, limitations or advantages of each approach along with dataset used for forgery detection.

3.4 Physical environment based technique

If images which are used to create a composite are taken in different lighting conditions, then it is hard to match their lighting conditions in the combined photograph. If there is any unrealistic lighting observed in the image, then it is possible that it has been tampered with. The first attempt was made by Johnson et al. They had taken the direction of the light source for the different objects and used it to detect the tampered region [67]. Later, they improved the method to include multiple light sources. This will work in complex lighting conditions [68]. Then the author, Johnson et al., introduced this in a 3-d direction [69] also, it detects forgeries with an accuracy of 83.7% [34]. A technique for detecting the 3-d lighting environment was suggested by Kee et al. [74]. Inconsistencies in the direction of the light source may be used to identify forgery in images, with a detection rate of 87.33% [106]. Fan et al. presented a technique for detecting the tampered region in 2-d lighting based on the object shape and the shade of the light [44]. De Carvalh et al. proposed a technique for detecting the counterfeit region in an image using discrepancies in color lighting and illumination based on statistical data. SVM and fusion classifiers are employed to identify forgeries, and the detection rate is 86% [40]. Table 5 presents a overview of comparison among physical environment based techniques in terms of approach/method adopted, type of forgery detected, limitations or advantages of each approach along with dataset used for forgery detection.

3.5 Geometry based technique

Geometry-based techniques are employed to measure the location, shape and size of objects in an image with respect to its camera position. Normally, the principal point is near the center of the image. If the person or the object gets trans-located in the image, proportionally, the principal point also moves. So the difference between principal points across the image is used for tampering detection. Johnson et al. introduced how to estimate the principal point from the image and other geometric shapes. They demonstrate how the translation plane is equal to the principal point shift and how these values can be used to detect tampering [70, 119].

4 Datasets

Datasets are the main source for testing the efficiency of different image forgery detection algorithms. Researchers have created few datasets and made them available publicly. These publicly available dataset help researcher to examine their method for detecting the tampered area in different scenario. ISCAS and Columbia datasets are used for splicing methods where ISCAS dataset contains about 20 images with size of 1024*1024–2600 *2000 [87] and Columbia contains about 1845 images with size of 128*128 pixels. These images are extracted from Calphotos collections and some other images are collected from digital cameras [133]. Then for copy-move technique DVMM dataset is used where DVMM constitute different

Table 4 A comparison among camera based techniques

Ref.No.	Method	Type of forgery detected	Remarks	Dataset
[46]	Ballistic technique	JPEG quantization and camera source	Ballistic technique is helpful in forensic examination	–
[75]	SVM	Identification source camera	Powerful features could be extracted to enhance the detection capability.	Own dataset
[21]	SVM	Based on color filter array camera source is identified.	This is limited to compressed image.	
[48]	Sensor noise and feature extraction methods	Sensor noise is detected	By combining the different methods helps to accurately detect the camera	Dresden
[97]	Feed forward network and back propagation network	Robust to JPEG compression, Gaussian noise, gamma correction and median filter smoothing	For better results 3 channels can be combined.	–
[141]	Convolutional neural network, ResNet50 network	Source camera identification	Improved camera model and source identification.	Dresden
[142]	Lens radial distortion	Source camera identification	Detection rate is less when zoom range is high	–

Table 5 A comparison among physical environment based techniques

Ref. No.	Method	Type of forgery detected	Remarks	Dataset
[67]	Infinite light source [2d and 3d]	Direction of lighting	Creation or removal of shadow is identified	–
[68]	Comparing lighting environments	3-d direction light	Ability to estimate complex lighting conditions.	–
[34]	Hestenes-Powell multiplier	Identify light source direction	Adaptive identification can be done further	–
[74]	3-D estimation, 3-D registration methods	3-D light direction is identified	Remove ambiguities than previous techniques.	–
[106]	Lambert illuminant model	Detection of 2-d light direction.	Higher true positive rate and less time complexity	–

datasets, first dataset from Columbia university in which image size of 128*128 pixels and second one is from kodak corporation which contain compressed PNG images [27, 135]. According to the type of manipulation, datasets are used. So CoMoFoD dataset is used for copy-move forgery detection, where the dataset is grouped according to the manipulation. CoMoFoD contains 250 forged images with 512*512 size [10]. Case of pixel based method clone, deletion, resampling is also main concern. To detect this type of manipulation Nimble or NIST-16 dataset is introduced where it contains different level of post-processing and compressed image versions. This Nimble dataset contains 2520 manipulated images [25, 109, 110].

For classifying tampered and non-tampered images, VOC2012 dataset is used. VOC2012 consists of untouched images which help to detect the splicing [92]. To achieve good result with low feature vector CASIA datasets are introduced. This CASIA is of 2 versions CASIA v1.0 and CASIA v2.0. CASIA v1.0 consists of 1721 images with 384*256 image size. All 1721 images are JPG format with 800 authentic images and 921 spliced images. CASIA v2.0 consists of 12,614 images with 240*160 image size. 12,614 images are JPG, TIFF and bmp format with 7491 authentic images and 5123 spliced images [13, 43, 109]. To detect the exact forged area MICC-F220 and MICC-F2000 datasets are used. MICC-F220 consists of 220 images divided into 110 forged images and 110 original images. 1.2% of the image area is tampered in the whole image. MICC-F2000 consists of 2000 images divided into 700 tampered images and 1300 original images. 1.12% of the image area is tampered in the whole image [9, 23, 99]. This dataset gives accurate result compare to other datasets.

5 Conclusion and future work

Image authentication is a process of enabling an image to verify its authenticity and integrity. With active image authentication (also known as explicit authentication), this has been accomplished effectively. The most compatible and scalable method is passive image authentication (also known as implicit authentication or automatic authentication), but it is often unable to make precise decisions about image integrity and authenticity, leading to the development of a vast field of study known as digital image forensics. In this paper, we presented a review of previous research on image authentication for tamper detection with localization. Specifically, a detailed survey has been carried out on active image authentication techniques meant for localization of tampered regions in the image. These techniques are compared in terms of their methodology (viz. watermarking, signature or hybrid), image authentication domain (viz. spatial or frequency), pixel based or block based, image recoverability and sensitive to malicious or non-malicious attacks. An overview of literature on passive image tampering detection with localization has also been presented.

In the same way that fake images deceive viewers, sophisticated image editing tools can deceive image forensic tools. Hence, in case of digital images, ‘protection is always better than detection’. Image authentication using active approach is not compatible and scalable in terms of implementation in the current scenario. Therefore, we need an intelligent authentication strategy which is a blend of explicit and automatic authentication that can be deployed easily without human intervention.

At first, multimedia data are usually précised, thereafter transferred and set aside. The hash which is developed at compression stage is highly reliable and can be applied easily compared to the hash of raw data. Hash generation for cipher image is more securable and maintains high confidentiality compare to raw image.

Code availability Not Availability.

Funding Not Applicable.

Data availability Not Applicable.

Declarations

Conflicts of interest/competing interests Not Applicable.

References

1. Abbas SQ, Ahmed F, Živić N, Ur-Rehman O (2016) Perceptual image hashing using SVD based noise resistant local binary pattern. In: 8th International Congress on Ultra Modern Telecommunications and Control Systems and Workshops, pp. 401–407
2. Abdalla Y, Iqbal MT, Shehata M (2019) Copy-move forgery detection and localization using a generative adversarial network and convolutional neural-network. Information 10(9):286
3. Abdelhakim A, Saleh HI, Abdelhakim M (2019) Fragile watermarking for image tamper detection and localization with effective recovery capability using K-means clustering. Multimed Tools Appl 78(22): 32523–32563
4. Abidin ABZ, Majid HBA, Samah ABA, Hashim HB (2019) Copy-move image forgery detection using deep learning methods: a review. In: 2019 6th international conference on research and innovation in information systems (ICRIIS), pp. 1–6
5. Adams J, Parulski K, Spaulding K (1998) Color processing in digital cameras. IEEE Micro 18(6):20–30
6. Ahmed F, Siyal MY (2007) A secure and robust wavelet-based hashing scheme for image authentication. In: International conference on multimedia modeling. Springer, Berlin, Heidelberg, pp. 51–62
7. Ahmed F, Siyal MY, Abbas VU (2010) A secure and robust hash-based scheme for image authentication. Signal Process 90(5):1456–1470
8. Ahmed B, Gulliver TA, alZahir S (2020) Image splicing detection using mask-RCNN. Signal Image Video Process 14(5):1035–1042
9. Al-Azrak F. M. , Sedik A, Dessowky MI, El Banby GM, Khalaf AA, Elkorany AS, El-Samie FEA (2020) An efficient method for image forgery detection based on trigonometric transforms and deep learning. Multimed Tools Appl 79(25):18221–18243
10. Alkawaz MH, Sulong G, Saba T, Rehman A (2018) Detection of copy-move image forgery based on discrete cosine transform. Neural Comput Appl 30(1):183–192
11. Amerini I, Ballan L, Caldelli R, Del Bimbo A, Del Tongo L, Serra G (2013) Copy-move forgery detection and localization by means of robust clustering with J-linkage. Signal Process Image Commun 28(6):659–669
12. Anbu T, Joe MM, Murugeswari G (2020) A comprehensive survey of detecting tampered images and localization of the tampered region. Multimedia Tools and Applications, pp. 1–39
13. Armas Vega EA, Sandoval Orozco AL, García Villalba LJ, Hernandez-Castro J (2018) Digital images authentication technique based on DWT, DCT and local binary patterns. Sensors 18(10):3372
14. Asghar K, Habib Z, Hussain M (2017) Copy-move and splicing image forgery detection and localization techniques: a review. Aust J Forensic Sci 49(3):281–307
15. Azeroual A, Afdel K (2017) Real-time image tamper localization based on fragile watermarking and Faber-Schauder wavelet. AEU-Int J Electron Commun 79:207–218
16. Babu B, Pillai KA (2015) Robust hashing for image authentication using Zernike moments, gabor wavelets and histogram features. Int J Sci Technol Manag 04(01)
17. Barad ZJ, Goswami MM (2020) Image forgery detection using deep learning: A survey. In: 2020 6th international conference on advanced computing and communication systems (ICACCS), pp. 571–576.
18. Bashar M, Noda K, Ohnishi N, Mori K (2010) Exploring duplicated regions in natural images. IEEE Trans Image Process:1–40
19. Battiato S, Farinella GM, Messina E, Puglisi G (2012) Robust image alignment for tampering detection. IEEE Trans Inf Forensics Secur 7(4):1105–1117
20. Battiato S, Farinella G. M, Messina E, Puglisi G (201) Understanding geometric manipulations of images through BOVW-based hashing. In: IEEE International Conference on Multimedia and Expo, pp. 1–6.

21. Bayram S, Sencar H, Memon N, Avcibas I (2005) Source camera identification based on CFA interpolation. In: IEEE International Conference on Image Processing, Vol. 3, pp. III-69
22. Bianchi T, Piva A (2012) Image forgery localization via block-grained analysis of JPEG artifacts. IEEE Trans Inf Forensics Secur 7(3):1003–1017
23. Bilal M, Habib HA, Mehmood Z, Saba T, Rashid M (2020) Single and multiple copy–move forgery detection and localization in digital images based on the sparsely encoded distinctive features and DBSCAN clustering. Arab J Sci Eng 45(4):2975–2992
24. Bolourian Haghighi B, Taherinia AH, Monsefi R (2020) An effective semi-fragile watermarking method for image authentication based on lifting wavelet transform and feed-forward neural network. Cognit Comput 12(4):863–890
25. Bunk J, Bappy JH, Mohammed TM, Nataraj L, Flenner A, Manjunath BS, Peterson L (2017) Detection and localization of image forgeries using resampling features and deep learning. In: 2017 IEEE conference on computer vision and pattern recognition workshops (CVPRW), pp. 1881-1889
26. Byun SW, Son HS, Lee SP (2019) Fast and robust watermarking method based on DCT specific location. IEEE Access 7:100706–100718
27. Cao Y, Gao T, Fan L, Yang Q (2012) A robust detection algorithm for copy-move forgery in digital images. Forensic Sci Int 214(1–3):33–43
28. Celik M. U, Sharma G, Saber E, Tekalp AM (2002) Hierarchical watermarking for secure image authentication with localization. IEEE Transaction Image Processing 11(6):585–595
29. Celik MU, Sharma G, Tekalp AM (2006) Lossless watermarking for image authentication: a new framework and an implementation. IEEE Trans Image Process 15(4):1042–1049
30. Chang CC, Hu YS, Lu TC (2006) A watermarking-based image ownership and tampering authentication scheme. Pattern Recogn Lett 27(5):439–446
31. Chen YL, Hsu CT (2011) Detecting recompression of JPEG images via periodicity analysis of compression artifacts for tampering detection. IEEE Trans Inf Forensics Secur 6(2):396–406
32. Chen T, Wang J, Zhou Y (2001) Combined digital signature and digital watermark scheme for image authentication. In: International Conferences on Info-Tech and Info-NetProceedings, Cat No 01EX479, Vol. 5, pp. 78–82
33. Chen W, Shi YQ, Su W (2007) Image splicing detection using 2D phase congruency and statistical moments of characteristic function. In: Security, Steganography, and Watermarking of Multimedia Contents IX, Vol. 6505, p. 65050R.
34. Chen H, Shen X, Lv Y (2010) Blind identification method for authenticity of infinite light source images. In: 2010 Fifth International Conference on Frontier of Computer Science and Technology. pp. 131–135
35. Chennamma HR, Rangarajan L (2011) Image splicing detection using inherent lens radial distortion. Int J Comput Sci Issues 7(6) 2010:149–158 ISSN.arXiv preprint arXiv:1105.4712
36. Cheung YM, Wu HT (2007) A sequential quantization strategy for data embedding and integrity verification. IEEE Trans Circuits Syst Video Technol 17(8):1007–1016
37. Ciptasari RW, Rhee KH, Sakurai K (2013) Exploiting reference images for image splicing verification. Digit Investig 10(3):246–258
38. Dadkhah S, Abd Manaf A, Sadeghi S (2014) Efficient image authentication and tamper localization algorithm using active watermarking. In: Bio-inspiring Cyber Security and Cloud Services: Trends and Innovations, pp 115–148
39. Davarzani R, Mozaffari S, Yaghmaie.K. (2015) Image authentication using LBP-based perceptual image hashing. Journal of AI and Data Mining 3(1):21–30
40. De Carvalho TJ, Riess C, Angelopoulou E, Pedrini H, de Rezende Rocha A (2013) Exposing digital image forgeries by illumination color classification. IEEE Trans Inf Forensics Secur 8(7):1182–1194
41. Ding W, Xie Y, Wang Y (2019) Image authentication and tamper localization based on relative difference between DCT coefficient and its estimated value. Multimed Tools Appl 78(5):5305–5328
42. Dirik AE, Sencar HT, Memon N (2007) Source camera identification based on sensor dust characteristics. In: 2007 IEEE Workshop on Signal Processing Applications for Public Security and Forensics, pp. 1–6
43. Doegar A, Dutta M, Kumar G (2020) Image forgery detection using Google net and random Forest machine learning algorithm. J Univ Shanghai Sci Technol:1271–1278
44. Fan W, Wang K, Cayre F, Xiong Z (2012) 3D lighting-based image forgery detection using shape-from-shading. In: 2012 Proceedings of the 20th European Signal Processing Conference, pp. 1777–1781
45. Fang Z, Wang S, Zhang X (2010) Image splicing detection using color edge inconsistency. In: 2010 International Conference on Multimedia Information Networking and Security, pp. 923–926
46. Farid H (2006) Digital image ballistics from JPEG quantization. Computer Science Technical Report TR2006–583. https://digitalcommons. (2006). dartmouth.edu/cs_tr/291
47. Farid H (2009) Image forgery detection. IEEE Signal Process Mag 26(2):16–25

48. Farinella GM, Giuffrida MV, Digiacomo V, Battiato S (2015) On blind source camera identification. In: International Conference on Advanced Concepts for Intelligent Vision Systems, pp. 464–473
49. Fridrich J (1998) Image watermarking for tamper detection. In: Proceedings 1998 International Conference on Image Processing, Cat. No. 98CB36269, Vol. 2, pp. 404–408
50. Fridrich J (1999) Robust bit extraction from images. In: Proceedings IEEE International Conference on Multimedia Computing and Systems Vol. 2, pp. 536–540
51. Fridrich AJ, Soukal BD, Lukáš AJ (2003) Detection of copy-move forgery in digital images. In: Proceedings of Digital Forensic Research Workshop
52. Fu D, Shi YQ, Su W (2006) Detection of image splicing based on Hilbert-Huang transform and moments of characteristic functions with wavelet decomposition. In: International workshop on digital watermarking, pp. 177–187
53. Fu D, Shi YQ, Su W (2007) A generalized Benford's law for JPEG coefficients and its applications in image forensics. In: Security, Steganography, and Watermarking of Multimedia Contents IX, Vol. 6505, p. 65051L
54. Ghorbani M, Firouzmand M, Faraahi A (2011) DWT-DCT (QCD) based copy-move image forgery detection. In: 2011 18th International Conference on Systems, Signals and Image Processing, IEEE, pp 1–4
55. Gill NK, Garg R, Doegar EA (2017) A review paper on digital image forgery detection techniques. In: 2017 8th international conference on computing, communication and networking technologies (ICCCNT), pp. 1–7
56. Gong D, Goh OS, Kumar YJ, Ye Z, Chi W (2020) Deepfake forensics, an ai-synthesized detection with deep convolutional generative adversarial networks. Int J 9(3):2861–2870
57. Gopi ES, Lakshmanan N, Gokul T, Kumara Ganesh S (2006) Digital image forgery detection using artificial neural network and auto regressive coefficients. In: 2006 Canadian Conference on Electrical and Computer Engineering, pp. 194–197
58. Guo XC, Hatzinakos D (2007) Content based image hashing via wavelet and radon transform. In: Pacific-rim conference on multimedia. Springer, Berlin, Heidelberg, pp. 755–764
59. Guorui F, Jian W (2020) Image forgery detection based on the convolutional neural network. In: proceedings of the 2020 12th international conference on machine learning and computing, pp. 266–270
60. Gupta M, Kishore RR (2021) A survey of watermarking technique using deep neural network architecture. In: 2021 international conference on computing, communication, and intelligent systems (ICCCIS), pp. 630–635
61. He Z, Sun W, Lu W, Lu H (2011) Digital image splicing detection based on approximate run length. Pattern Recogn Lett 32(12):1591–1597
62. He Z, Lu W, Sun W, Huang J (2012) Digital image splicing detection based on Markov features in DCT and DWT domain. Pattern Recogn 45(12):4292–4299
63. Hosseini S, Mahdavi M (2021) Image content dependent semi-fragile watermarking with localized tamper detection. arXiv preprint arXiv:2106.14150
64. Huang H, Guo W, Zhang Y (2008) Detection of copy-move forgery in digital images using SIFT algorithm. In: 2008 IEEE Pacific-Asia Workshop on Computational Intelligence and Industrial Application, IEEE, Vol 2, pp 272–276
65. Huang Y, Lu W, Sun W, Long D (2011) Improved DCT-based detection of copy-move forgery in images. Forensic Sci Int 206(1–3):178–184
66. Islam MM, Karmakar G, Kamruzzaman J, Murshed M, Kahandawa G, Parvin N (2018) Detecting splicing and copy-move attacks in color images. In: 2018 Digital Image Computing: Techniques and Applications. pp. 1–7
67. Johnson MK, Farid H (2005) Exposing digital forgeries by detecting inconsistencies in lighting. In: Proceedings of the 7th workshop on Multimedia and security, pp 1–10
68. Johnson MK, Farid H (2007) Exposing digital forgeries in complex lighting environments. IEEE Trans Inf Forensics Secur 2(3):450–461
69. Johnson MK, Farid H (2007) Exposing digital forgeries through specular highlights on the eye. In: International Workshop on Information Hiding, pp. 311–325
70. Johnson MK, Farid H (2007) Detecting photographic composites of people. In: International Workshop on Digital Watermarking, pp. 19–33
71. Kakar P, Sudha N (2012) Exposing postprocessed copy–paste forgeries through transform-invariant features. IEEE Trans Inf Forensics Secur 7(3):1018–1028
72. Kamili A, Hurrah NN, Parah SA, Bhat GM, Muhammad K (2020) DWFCAT: dual watermarking framework for industrial image authentication and tamper localization. IEEE Trans Ind Inf 17(7):5108–5117
73. Kang X, Wei S (2008) Identifying tampered regions using singular value decomposition in digital image forensics. In: 2008 International conference on computer science and software engineering, IEEE, Vol 3, pp 926–930

74. Kee E, Farid H (2010) Exposing digital forgeries from 3-D lighting environments. In: 2010 IEEE International Workshop on Information Forensics and Security, pp. 1–6
75. Kharrazi M, Sencar HT, Memon N (2004) Blind source camera identification. In: 2004 International Conference on Image Processing, Vol. 1, pp. 709–712
76. Kharrazi M, Sencar H, Memon N (2005) Blind source camera identification. In: IEEE International Conference on Image Processing, pp. 69–72
77. Kim C, Yang CN (2021) Self-embedding fragile watermarking scheme to detect image tampering using AMBTC and OPAP approaches. Appl Sci 11(3):1146
78. Kirchner M (2008) Fast and reliable resampling detection by spectral analysis of fixed linear predictor residue. In: Proceedings of the 10th ACM workshop on Multimedia and security, pp. 11–20
79. Kundur D, Hatzinakos D (1998) Towards a telltale watermarking technique for tamper-proofing. In: Proceedings 1998 International Conference on Image Processing, Cat No 98CB36269, Vol. 2, pp. 409–413
80. Langille A, Gong M (2006) An efficient match-based duplication detection algorithm. In: The 3rd Canadian Conference on Computer and Robot Vision, CRV'06, pp. 64–64
81. Laouamer L, AlShaikh M, Nana L, Pascu AC (2015) Robust watermarking scheme and tamper detection based on threshold versus intensity. J Innov Digital Ecosyst 2(1–2):1–12
82. Li H, Huang J (2019) Localization of deep inpainting using high-pass fully convolutional network. In proceedings of the IEEE/CVF international conference on computer vision, pp. 8301–8310
83. Li G, Wu Q, Tu D, Sun S (2007) A sorted neighborhood approach for detecting duplicated regions in image forgeries based on DWT and SVD. In: 2007 IEEE international conference on multimedia and expo, pp. 1750–1753
84. Li B, Shi YQ, Huang J (2008) Detecting doubly compressed JPEG images by using mode based first digit features. In: 2008 IEEE 10th Workshop on Multimedia Signal Processing, pp. 730–735
85. Li X, Jing T, Li X (2010) Image splicing detection based on moment features and Hilbert-Huang Transform. In: 2010 IEEE international conference on information theory and information security, pp. 1127–1130
86. Lin CY, Chang SF (2000) Semi-fragile watermarking for authenticating JPEG visual content. In Security and Watermarking of Multimedia Contents II, International Society for Optics and Photonics, Vol. 3971, pp. 140–151
87. Lin SD, Wu T (2011) An integrated technique for splicing and copy-move forgery image detection. In: 2011 4th International Congress on Image and Signal Processing, Vol. 2, pp. 1086–1090
88. Lin HY, Liao HY, Lu CS, Lin JC (2005) Fragile watermarking for authenticating 3-D polygonal meshes. IEEE Trans Multimed 7(6):997–1006
89. Lin HJ, Wang CW, Kao YT (2009) Fast copy-move forgery detection. WSEAS Trans Signal Process 5(5): 188–197
90. Lin HJ, Wang CW, Kao YT, Chen S (2009) An efficient method for copy-move forgery detection. In WSEAS international conference. Proceedings. Mathematics and computers in science and engineering (No. 8). World Scientific and Engineering Academy and Society
91. Lin Z, He J, Tang X, Tang CK (2009) Fast, automatic and fine-grained tampered JPEG image detection via DCT coefficient analysis. Pattern Recogn 42(11):2492–2501
92. Liu B, Pun CM (2018) Deep fusion network for splicing forgery localization. In proceedings of the European conference on computer vision (ECCV) workshops, pp. 0-0
93. Liu Q, Sung AH (2009) A new approach for JPEG resize and image splicing detection. In: Proceedings of the First ACM workshop on Multimedia in forensics, pp. 43–48
94. Liu G, Wang J, Lian S, Wang Z (2011) A passive image authentication scheme for detecting region-duplication forgery with rotation. J Netw Comput Appl 34(5):1557–1565
95. Liu Q, Cao X, Deng C, Guo X (2011) Identifying image composites through shadow matte consistency. IEEE Trans Inf Forensics Secur 6(3):1111–1122
96. Liu X, Liang J, Wang ZY, Tsai YT, Lin CC, Chen CC (2020) Content-based image copy detection using convolutional neural network. Electronics 9(12):2029
97. Long Y, Huang Y (2006) Image based source camera identification using demosaicking. In: 2006 IEEE Workshop on Multimedia Signal Processing, pp. 419–424
98. Lou DC, Liu JL (2000) Fault resilient and compression tolerant digital signature for image authentication. IEEE Trans Consum Electron 46(1):31–39
99. Lourembam A, Kumar KMVM, Singh TR (2021) A robust image copy detection method using machine learning. Malaya J Mat:23–30
100. Lu CS, Liao HY (2001) Multipurpose watermarking for image authentication and protection. IEEE Trans Image Process 10(10):1579–1592
101. Lu W, Wu M (2010) Multimedia forensic hash based on visual words. In: 2010 IEEE International Conference on Image Processing, pp. 989–992

102. Lu W, Varna AL, Wu M (2010) Forensic hash for multimedia information. In: Media Forensics and Security II. International Society for Optics and Photonics. Vol. 7541, p. 75410Y
103. Lukáš J, Fridrich J (2003) Estimation of primary quantization matrix in double compressed JPEG images. In: Proc. Digital forensic research workshop, pp. 5–8
104. Luo. W, Huang. J, Qiu. G (2006) Robust detection of region-duplication forgery in digital image. In: 18th International Conference on Pattern Recognition, Vol. 4, pp. 746–749
105. Lv X, Wang ZJ (2012) Perceptual image hashing based on shape contexts and local feature points. IEEE Trans Inf Forensics Secur 7(3):1081–1093
106. Lv Y, Shen X, Chen H (2011) An improved image blind identification based on inconsistency in light source direction. J Supercomput 58(1):50–67
107. Lynch G, Shih FY, Liao HYM (2013) An efficient expanding block algorithm for image copy-move forgery detection. Inf Sci 239:253–265
108. Mahdian B, Saic S (2008) Blind authentication using periodic properties of interpolation. IEEE Trans Inf Forensics Secur 3(3):529–538
109. Mazaheri G, Mithun NC, Bappy JH, Roy-Chowdhury AK (2019). A skip connection architecture for localization of image manipulations. In CVPR Workshops, pp 119–129
110. Mohammed TM, Bunk J, Nataraj L, Bappy JH, Flenner A, Manjunath BS, Chandrasekaran S, Roy-Chowdhury AK, Peterson LA (2018) Boosting image forgery detection using resampling features and copy-move analysis. Electronic Imaging 2018(7):118–111
111. Molina-Garcia J, Garcia-Salgado BP, Ponomaryov V, Reyes-Reyes R, Sadovnychiy S, Cruz-Ramos C (2020) An effective fragile watermarking scheme for color image tampering detection and self-recovery. Signal Process Image Commun 81:115725
112. Muhammad G, Hussain M, Bebis G (2012) Passive copy move image forgery detection using undecimated dyadic wavelet transform. Digit Investig 9(1):49–57
113. Muzaffer G, Ulutas G (2019) A new deep learning-based method to detection of copy-move forgery in digital images. In: 2019 scientific meeting on Electrical-Electronics & Biomedical Engineering and computer science (EBBT), pp. 1-4
114. Myna AN, Venkateshmurthy MG, Patil CG (2007) Detection of region duplication forgery in digital images using wavelets and log-polar mapping. In: International Conference on Computational Intelligence and Multimedia Applications, Vol. 3, pp. 371–377.
115. Ng TT (2009) Camera response function signature for digital forensics-part II: signature extraction. In: 2009 First IEEE International Workshop on Information Forensics and Security, pp. 161–165
116. Ng TT, Chang SF (2004) A model for image splicing. In 2004 International Conference on Image Processing, Vol. 2, pp. 1169–1172
117. Ng TT, Tsui MP (2009) Camera response function signature for digital forensics-part I: theory and data selection
118. Ng TT, Chang SF, Sun Q (2004) Blind detection of photomontage using higher order statistics. In: 2004 IEEE international symposium on circuits and systems, Cat. No. 04CH37512, Vol. 5, pp. V-V
119. Ng TT, Chang SF, Hsu J, Xie L, Tsui MP (2005) Physics-motivated features for distinguishing photographic images and computer graphics. In: Proceedings of the 13th annual ACM international conference on Multimedia, pp. 239–248
120. Nguyen TT, Nguyen CM, Nguyen DT, Nguyen DT, Nahavandi S (2019) Deep learning for deepfakes creation and detection: A survey. arXiv preprint arXiv:1909.11573
121. O'Ruanaidh JJ, Pun T (1997) Rotation, scale and translation invariant digital image watermarking In: Proceedings of International Conference on Image Processing, Vol. 1, pp. 536–539
122. Pal P, Jana B, Bhaumik J (2021) An image authentication and tampered detection scheme exploiting local binary pattern along with hamming error correcting code. Wirel Pers Commun 121(1):939–961
123. Pan X, Lyu S (2010) Region duplication detection using image feature matching. IEEE Trans Inf Forensics Secur 5(4):857–867
124. Paquet AH, Ward RK (2002) Wavelet-based digital watermarking for image authentication. In IEEE CCECE2002Canadian Conference on Electrical and Computer Engineering, Conference Proceedings, Cat No 02CH37373, Vol. 2, pp. 879–884
125. Pevny T, Fridrich J (2008) Detection of double-compression in JPEG images for applications in steganography. IEEE Trans Inf Forensics Secur 3(2):247–258
126. Pickett JP (2000) The American heritage dictionary, 4th edn. Boston, Massachusetts, Houghton Mifflin Company
127. Popescu AC, Farid H (2004) Exposing digital forgeries by detecting duplicated image regions. Computer Science Technical Report TR2004–515. 8-1-2004. https://digitalcommons.dartmouth.edu/cs_tr/254
128. Popescu A. C, Farid H (2004) Statistical tools for digital forensics. In: International workshop on information hiding, pp. 128–147

129. Popescu AC, Farid H (2005) Exposing digital forgeries by detecting traces of resampling. IEEE Trans Signal Process 53(2):758–767
130. Prasad S, Ramakrishnan KR (2006) On resampling detection and its application to detect image tampering. In: 2006 IEEE International Conference on Multimedia and Expo, pp. 1325–1328
131. Preda RO (2013) Semi-fragile watermarking for image authentication with sensitive tamper localization in the wavelet domain. Measurement 46(1):367–373
132. Pun CM, Yan CP, Yuan XC (2018) Robust image hashing using progressive feature selection for tampering detection. Multimed Tools Appl 77(10):11609–11633
133. Qu Z, Qiu G, Huang J (2009) Detect digital image splicing with visual cues. In: International workshop on information hiding, pp. 247–261
134. Ranjbar S, Zargari F, Ghanbari M (2013) A highly robust two-stage contourlet-based digital image watermarking method. Signal Process Image Commun 28(10):1526–1536
135. Rao Y, Ni J (2016) A deep learning approach to detection of splicing and copy-move forgeries in images. In: 2016 IEEE international workshop on information forensics and security (WIFS) pp. 1-6
136. Rao MP, Rajagopalan AN, Seetharaman G (2014) Harnessing motion blur to unveil splicing. IEEE Trans Inf Forensics Secur 9(4):583–595
137. Roy S, Sun Q (2007) Robust hash for detecting and localizing image tampering. In: 2007 IEEE International Conference on Image Processing. Vol. 6, pp. VI-117
138. Roy S, Sun Q (2007) Robust hash for detecting and localizing image tampering. In: IEEE International Conference on Image Processing, Vol. 6, pp. VI-117
139. Sadeghi S, Dadkhah S, Jalab HA, Mazzola G, Uliyan D (2018) State of the art in passive digital image forgery detection: copy-move image forgery. Pattern Anal Applic 21(2):291–306
140. Sameer VU, Sugumaran S, Naskar R (2018) K-unknown models detection through clustering in blind source camera identification. IET Image Process 12(7):1204–1213
141. Sameer VU, Dali I, Naskar R (2018) A deep learning based digital forensic solution to blind source identification of Facebook images. In: International Conference on Information Systems Security, pp. 291–303
142. San Choi K, Lam EY, Wong KK (2006) Source camera identification using footprints from lens aberration. In: Digital Photography II, Vol. 6069, p. 60690J
143. Sekeh MA, Maarof MA, Rohani MF, Motiei M (2011) Sequential straightforward clustering for local image block matching. World Acad Sci Eng Technol 50:774–778
144. Sekeh MA, Maarof MA, Rohani MF, Mahdian B (2013) Efficient image duplicated region detection model using sequential block clustering. Digit Investig 10(1):73–84
145. Shao H, Yu T, Xu M, Cui W (2012) Image region duplication detection based on circular window expansion and phase correlation. Forensic Sci Int 222(1–3):71–82
146. Shin J, Kim D, Ruland C (2014) Content based image authentication using HOG feature descriptor. In: IEEE International Conference on Image Processing. pp. 5292–5296
147. Singh OP, Singh AK, Srivastava G, Kumar N (2021) Image watermarking using soft computing techniques: A comprehensive survey. Multimed Tools Appl 80(20):30367–30398
148. Sumalatha L, Krishna VV, Babu AV (2012) Image content authentication based on wavelet edge features. Int J Comput Appl 49(23):24–29
149. Sutthiwan P, Shi YQ, Su W, Ng TT (2010) Rake transform and edge statistics for image forgery detection. In: 2010 IEEE International Conference on Multimedia and Expo, pp. 1463–1468
150. Tagliasacchi M, Valenzise G, Tubaro S (2009) Hash-based identification of sparse image tampering. IEEE Trans Image Process 18(11):2491–2504
151. Tang S, Li JT, Zhang YD (2005) Compact and robust image hashing. In: International Conference on Computational Science and Its Application. Springer, Berlin, Heidelberg, pp. 547–556
152. Verdoliva L (2020) Media forensics and deepfakes: an overview. IEEE J Select Topics Signal Process 14(5):910–932
153. Vinolin V, Sucharitha M (2021) Hierarchical categorization and review of recent techniques on image forgery detection. Comput J 64(11):1692–1704
154. Wang J, Li Y (2019) Splicing image and its localization: a survey. Journal of Information Hiding and Privacy Protection 1(2):77–86
155. Wang W, Dong J, Tan T (2009) Effective image splicing detection based on image chroma. In: 2009 16th IEEE international conference on image processing, IEEE, pp 1257–1260
156. Wang X, Zheng N, Xue J, Liu Z (2012) A novel image signature method for content authentication. Comput J 55(6):686–701
157. Wang X, Xue J, Zheng Z, Liu Z, Li N (2012) Image forensic signature for content authenticity analysis. J Vis Commun Image Represent 23(5):782–797
158. Wang W, Dong J, Tan T (2014) Exploring DCT coefficient quantization effects for local tampering detection. IEEE Trans Inf Forensics Secur 9(10):1653–1666

159. Wang X, Pang K, Zhou X, Zhou Y, Li L, Xue J (2015) A visual model-based perceptual image hash for content authentication. IEEE Trans Inf Forensics Secur 10(7):1336–1349
160. Wang X, Wang H, Niu S, Zhang J (2019) Detection and localization of image forgeries using improved mask regional convolutional neural network. Math Biosci Eng 16(5):4581–4593
161. Wang X, Zhou X, Zhang Q, Xu B, Xue J (2020) Image alignment based perceptual image hash for content authentication. Signal Process Image Commun 80:115642
162. Weng L, Braeckman G, Dooms A, Preneel B, Schelkens P (2012) Robust image content authentication with tamper location. In: 2012 IEEE international conference on multimedia and expo, pp. 380–385
163. Wolfgang RB, Delp EJ (1996) A watermark for digital images. In: Proceedings of 3rd IEEE International Conference on Image Processing, Vol. 3, pp. 219–222
164. Wolfgang RB, Delp EJ III (1999) Fragile watermarking using the VW2D watermark. In: Security and Watermarking of Multimedia Contents, vol 3657, pp 204–213
165. Wong PW (1998) A watermark for image integrity and ownership verification. In: PICS pp 374–379.
166. Wong PW, Memon N (2001) Secret and public key image watermarking schemes for image authentication and ownership verification. IEEE Trans Image Process 10(10):1593–1601
167. Wu X, Fang Z (2011) Image splicing detection using illuminantcolor inconsistency. In: 2011 Third International Conference on Multimedia Information Networking and Security, pp. 600–603
168. Wu M, Liu B (1998) Watermarking for image authentication. In: Proceedings 1998 International Conference on Image Processing Cat No 98CB36269, Vol. 2, pp. 437–441
169. Wu HC, Fan WL, Tsai CS, Ying JJC (2021) An image authentication and recovery system based on discrete wavelet transform and convolutional neural networks. Multimed Tools Appl 1-25
170. Yan CP, Pun CM, Yuan XC (2016) Quaternion-based image hashing for adaptive tampering localization. IEEE Trans Inf Forensics Secur 11(12):2664–2677
171. Yan CP, Pun CM, Yuan XC (2016) Multi-scale image hashing using adaptive local feature extraction for robust tampering detection. Signal Process 121:1–16
172. Yeo BL, Yeung MM (1999) Watermarking 3D objects for verification. IEEE Comput Graph Appl 19(1): 36–45
173. Yeung MM, Mintzer F (1997) An invisible watermarking technique for image verification. In: Proceedings of international conference on image processing. Vol. 2, pp. 680–683
174. Yeung MM, Mintzer F (1997) f watermarking technique for image verification. In: Proceedings of international conference on image processing, Vol. 2, pp. 680–683
175. Ying Q, Qian Z, Zhou H, Xu H, Zhang X, Li S (2021) From image to imuge: immunized image generation. In: Proceedings of the 29th ACM international conference on multimedia, pp. 3565-3573
176. Zhang J, Feng Z, Su Y (2008) A new approach for detecting copy-move forgery in digital images. In: 2008 11th IEEE Singapore International Conference on Communication Systems, pp. 362–366
177. Zhang Z, Kang J, Ren Y (2008) An effective algorithm of image splicing detection. In: 2008 international conference on computer science and software engineering, Vol. 1, pp. 1035–1039
178. Zhang W, Cao X, Qu Y, Hou Y, Zhao H, Zhang C (2010) Detecting and extracting the photo composites using planar homography and graph cut. IEEE Trans Inf Forensics Secur 5(3):544–555
179. Zhang H, Wang C, Zhou X (2017) Fragile watermarking based on LBP for blind tamper detection in images. J Inf Process Syst 13(2):385–399
180. Zhang H, Wang C, Zhou X (2017) Fragile watermarking for image authentication using the characteristic of SVD. Algorithms 10(1):27
181. Zhao X, Bateman P, Ho ATS (2011) Image authentication using active watermarking and passive forensics techniques. In: Multimedia Analysis, Processing and Communications. Springer, Berlin, pp 139–183
182. Zhao J, Guo J (2013) Passive forensics for copy-move image forgery using a method based on DCT and SVD. Forensic Sci Int 233(1–3):158–166
183. Zhao Y, Wang S, Feng G, Tang Z (2010) A robust image hashing method based on Zernike moments. J Comput Inf Syst 6(3):717–725
184. Zhao X, Li J, Li S, Wang S (2010) Detecting digital image splicing in chroma spaces. In: International workshop on digital watermarking, pp. 12–22
185. Zhao Y, Wang S, Zhang X, Yao H (2012) Robust hashing for image authentication using Zernike moments and local features. IEEE Trans Inf Forensics Secur 8(1):55–63

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.